# DEPARTMENT OF STATISTICS

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**To:**

The Editorial Board
PLOS Computational Biology
Public Library of Science

**Subject: Submission of Original Research Article for Consideration in PLOS Computational Biology**

Dear Editors and Editorial Board Members,

On behalf of all authors, I am pleased to submit our original research manuscript entitled "Integrating CRISPRi Perturbation Responses and Transcription-Factor Binding Priors for Regulatory Network Inference in Triple-Negative Breast Cancer" by Omar Faroque, Mst. Homayra Jannaty, Shahriar Hasan Abir, and Suman Biswas, for consideration for publication as a Research Article in PLOS Computational Biology under the Genomics, Epigenomics, & Proteomics section (with Systems Biology as secondary scope).

**1. Why is this manuscript suitable for publication in PLOS Computational Biology?**

Inferring faithful, directed gene regulatory networks (GRNs) in cancer systems biology is severely bottlenecked by the observational nature of single-cell transcriptomics, where pervasive dropout zero-inflation and indirect correlation confounding cause false discovery rates (FDR) exceeding 30–40%. Our manuscript introduces a computationally rigorous, interventional network inference framework that directly couples large-scale single-cell CRISPR-interference perturbation sequencing (Perturb-seq; GSE224986, comprising 95,678 cells across 11,975 sgRNAs) with physical ENCODE Factorbook ChIP-seq binding priors across 819 somatic mutation-associated super-enhancers in human triple-negative breast cancer (TNBC; MDA-MB-231). By applying Judea Pearl's do-calculus and explainable AI (SHAP TreeExplainer), our framework decodes non-linear threshold dynamics governing oncogenic master regulators (AP-1, MYC, EP300). Rigorous benchmarking against held-out experimental CRISPRi ground truth demonstrated strong discriminatory accuracy (AUROC = 0.8999, AUPRC = 0.8745, Precision = 90.13%, FDR = 9.87%), substantially outperforming observational co-expression and GENIE3. Furthermore, in silico driver feature silencing correlated with experimental single-cell knockdown fold changes ($R^2 = 0.473$, $p < 10^{-122}$), and prioritized master drivers significantly stratified overall patient survival across primary clinical cohorts (METABRIC n=320, TCGA-BRCA n=160; $p \leq 0.019$). This deep integration of mathematical causality, single-cell functional genomics, and clinical translatability directly aligns with the mission of PLOS Computational Biology.

**2. Why will our study inspire other members of the field, and how will it drive research forward?**

Our investigation bridges the methodological divide between high-throughput perturbation screens and directed transcriptional network modeling. First, we provide an empirical roadmap demonstrating how functional perturbation responses resolve the chronic high FDR of purely observational GRNs. Second, by moving beyond linear assumptions, our explainable ML framework demonstrates that oncogenic master regulators operate via steep sigmoidal transactivation thresholds, explaining why standard correlational methods fail to capture their regulatory influence. Third, our in silico therapeutic perturbation modeling reveals that while standard cytotoxic chemotherapy achieves only 38% topological network disruption, synergistic multi-target epigenetic combinations (BETi + HATi + HDACi) achieve up to 93% disruption of core oncogenic connectivity. This nominates rational, testable epigenetic combinations to overcome chemotherapy resistance in TNBC.

**Justification for Section Placement (Materials and Methods before Results):**

In accordance with PLOS Computational Biology instructions, we explicitly justify placing the Materials and Methods section immediately preceding Results. Because our study establishes a multi-layered mathematical and statistical workflow (integrating CRISPRi guide calling, Welch's interventional delta-testing, ChIP-seq prior mapping, PageRank directed multigraph centrality, and SHAP non-linear attribution), readers require an understanding of the formal pipeline to properly interpret the tripartite network decomposition, benchmarking metrics, and clinical prognostic models.

**Declarations, Data Availability & Fee Waiver:**

• Originality: This manuscript is an original work that has not been published previously and is not under consideration for publication elsewhere.
• Authorship: All co-authors have reviewed, contributed to, and approved the submitted version.
• Open Data & Code: In compliance with PLOS Open Science policies, all raw data are available via public accessions (NCBI GEO GSE224986, GSE223812; UCSC Factorbook). All processed datasets, reproducible Jupyter notebooks, and custom Python analysis scripts are freely available in our public GitHub repository: https://github.com/Omarrfaroqueiu79/tnbc-causal-grn under the open-source MIT license.
• Financial Disclosure & APC Waiver: The authors received no specific external funding for this work. As the corresponding author is affiliated with an institution in Bangladesh (Islamic University), this manuscript qualifies for an automatic 100% full Article Processing Charge (APC) waiver under the PLOS Global Participation Initiative (Research4Life Group A).

**Suggested Academic Editors and Expert Reviewers:**

We respectfully suggest the following independent researchers with expertise in single-cell perturbation genomics, causal network inference, and cancer epigenomics:
1. Dr. Nir Yosef, Department of Systems Immunology, Weizmann Institute of Science (Expertise: Single-cell GRNs, probabilistic models, perturbation genomics).
2. Dr. Dana Pe'er, Computational and Systems Biology Program, Memorial Sloan Kettering Cancer Center (Expertise: Single-cell causal inference, trajectory modeling).
3. Dr. Christina Leslie, Computational Biology Program, Memorial Sloan Kettering Cancer Center (Expertise: Epigenetic regulatory genomics, ChIP-seq prior integration).
4. Dr. Fabian Theis, Institute of Computational Biology, Helmholtz Munich (Expertise: Perturbation modeling, single-cell deep learning, Perturb-seq analysis).

Thank you for your consideration of our manuscript.

Sincerely,

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